4.1, 4.3, 4.5, 4.7, 4.9, 4.13, 4.15, 4.17

4.1 Show by induction that $2^n < n!$ for all $n \geq 4$.

For the base statement, n = 4, the statement that $2^4 < (4)!$ holds true as it boils down to $16 < 24$.
An inductive proof for this could be shown below
$2^{n+1} < (n+1)!$ which can be rewritten as $2(2^n) < (n+1)!$
Since $n > 4$, then $n + 1 > 5$, which means that $n + 1 > 2$ also hold true, multiplying $n!$ results in: $2n! < (n+1)n!$
If you multiply the induction hypothesis ($2^n < n!$) by 2, you get:
$2(2^n) < 2n!$
Then combining the above inequalities yields:
$2^{n+1} < 2n! < (n+1)!$
This proves that $2^n < n!$ holds for n+1. □

4.3 Show by induction that $f_0 + ... + f_n = f_{n+2} - 1$ in the Fibonacci sequence.

First, proving the base case:
For $n = 0$, $f_n = 1$ so $f_{n+2} - 1 = 2 - 1 = 1$ This proves the base case.
Now, assuming that n is true:
$\sum_{x=0}^{n} f_x = f_{n+2} - 1$
Finally, the inductive step:
so $f_{n+1} = f_{n+2} - 1$
$\sum_{x=0}^{n+1} f_x = f_{(n+1)+2} - 1 = f_{n+3} - 1.$
$\sum_{x=0}^{n+1} f_x = (\sum_{x=0}^{n} f_x) + f_{n+1}$
Using the induction hypothesis, the first part can be rewritten as: $\sum_{x=0}^{n+1} f_x = (f_{n+2} - 1) + f_{n+1} = (f_{n+2} + f_{n+1}) - 1$
Combining all of these here, $f_{n+2} + f_{n+1} = f_{n+3}$ so,
$\sum_{x=0}^{n+1} f_x = f_{n+3} - 1.$

4.5 Consider an alternative Fibonacci sequence, starting with 0, 2. Can you prove the analogue of theorem 25? Generalize the result as far as you can.

Theorem 25. For the standard Fibonacci sequence with $f_0 = 0$, $f_1 = 1$, and $f_{n+1} = f_n + f_{n-1}$ for $n \geq 1$, and for all integers $n \geq 0$, $\sum_{x=0}^{n} f_x^2 = f_n f_{n+1}$.

Generalization. Let f_n be any sequence satisfying $f_{n+1} = f_n + f_{n-1}$ for $n \geq 1$ with initial terms f_0, f_1. Then for all $n \geq 0$, $\sum_{x=0}^{n} f_x^2 = f_n f_{n+1} - f_0(f_1 - f_0)$.

Proof. We proceed by mathematical induction on n.

Base Case ($n = 0$): LHS $= \sum_{x=0}^{0} f_x^2 = f_0^2$
RHS $= f_0 f_1 - f_0(f_1 - f_0) = f_0 f_1 - f_0 f_1 + f_0^2 = f_0^2$
Since LHS = RHS, the base case holds.

Inductive Hypothesis: Assume the formula holds for an arbitrary integer $n \geq 0$: $\sum_{x=0}^{n} f_x^2 = f_n f_{n+1} - f_0(f_1 - f_0)$.

Inductive Step: We want to show that the formula holds for $n + 1$: $\sum_{x=0}^{n+1} f_x^2 = f_{n+1} f_{n+2} - f_0(f_1 - f_0)$.

Starting from the left-hand side: $\sum_{x=0}^{n+1} f_x^2 = (\sum_{x=0}^{n} f_x^2) + f_{n+1}^2$
Applying the inductive hypothesis: $\sum_{x=0}^{n+1} f_x^2 = \big(f_n f_{n+1} - f_0(f_1 - f_0)\big) + f_{n+1}^2$
Group the terms with f_{n+1}: $\sum_{x=0}^{n+1} f_x^2 = (f_n f_{n+1} + f_{n+1}^2) - f_0(f_1 - f_0)$
Factor out f_{n+1}: $\sum_{x=0}^{n+1} f_x^2 = f_{n+1}(f_n + f_{n+1}) - f_0(f_1 - f_0)$
Since $f_n + f_{n+1} = f_{n+2}$ by definition: $\sum_{x=0}^{n+1} f_x^2 = f_{n+1} f_{n+2} - f_0(f_1 - f_0)$.

4.7 Show by induction that a finite set with n elements has exactly 2^n many subsets.

Proposition. A finite set with n elements has exactly 2^n subsets for all integers $n \geq 0$.

Proof. We proceed by mathematical induction on n.

Base Case ($n = 0$): A set with 0 elements is the empty set, $\emptyset$. The only subset of $\emptyset$ is $\emptyset$ itself, which means it has 1 subset. Since $2^0 = 1$, the base case holds.

Inductive Hypothesis: Assume that any finite set with n elements has exactly 2^n subsets for some integer $n \geq 0$.

Inductive Step: Let S be a set with $n + 1$ elements. Choose an element $a \in S$, and define the set $S' = S \setminus \{a\}$.

The set S' contains n elements. By the inductive hypothesis, S' has 2^n subsets.

We can partition all subsets of S into two disjoint categories:
1. Subsets that do not contain a: These are precisely the subsets of S', so there are 2^n such subsets.
2. Subsets that contain a: Each of these subsets is uniquely formed by taking a subset $T \subseteq S'$ and forming $T \cup \{a\}$. Thus, there are also 2^n such subsets.

Summing the two categories gives the total number of subsets of S: $2^n + 2^n = 2 \cdot 2^n = 2^{n+1}$.

4.9 Prove that $\frac{1}{1*2} + \frac{1}{2*3} + \frac{1}{3*4} + \ldots + \frac{n}{n*n+1} = \frac{n}{n+1}$. Base case: $\sum_{x=1}^{n+1} f_x = (f_{n+2} - 1) + f_{n+1} = (f_{n+2} + f_{n+1}) - 1$

4.13 True or false: every flying polka-dotted elephant in this room is smoking a cigar. True! This holds true even though there isn't a polka-dotted elephant in the room, if there were, we don't know if the elephant would be smoking a cigar or not.

Criticize the following "proof."

Claim. $\sum_{i=1}^{\infty} \frac{1}{i} < \infty$.

Proof. **Let $S(n) = \sum_{i=1}^{n} \frac{1}{i}$, and consider the statement asserting $S(n) < \infty$. This statement is true for $n = 1$, since $\frac{1}{1} = 1 < \infty$. And if it is true for n, then it is true for $n + 1$, since the next sum $S(n + 1) = \sum_{i=1}^{n+1} \frac{1}{i}$ is equal to the previous sum $\sum_{i=1}^{n} \frac{1}{i}$ plus the next term $\frac{1}{n+1}$, or in other words, $S(n + 1) = S(n) + \frac{1}{n+1}$, and this is the sum of two finite numbers. So $S(1)$ is true, and $S(n)$ implies $S(n + 1)$. So we have therefore proved $\sum_{i=1}^{\infty} \frac{1}{i} < \infty$ by induction.** □

4.17 Prove that the least-number principle, the common induction principle, and the strong induction principle are all equivalent. Assuming any one of them, you can prove the others.